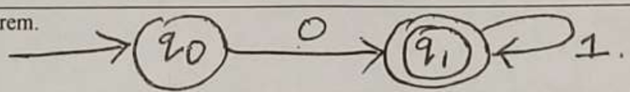
Internal Assessment – I

TERM : 15/4/24 TO 22/7/24	COURSE NAME: Theory Of Computations
DATE : 15/07/24 TIME:9:30 TO 10:30	COURSE CODE: AD44
MAX MARKS: 30	PORTIONS : L1 – L 15

Instructions to Candidates: **Answer any two full questions.** Marks: **15x2=30**

<table>
<tr><th>Q. NO</th><th>Questions</th><th>Blooms Levels (L1 to L6)*</th><th>CO</th><th>Marks</th></tr>
<tr><td>1.a</td><td>Design a DFA to accept strings having atleast two 0's in five length block over {0,1}.</td><td>L2</td><td>CO1</td><td>5</td></tr>
<tr><td>b</td><td>Consider the following ∈ - NFA.
i. Compute the ∈- closure of each state.
ii. Convert the automata to DFA.
<table>
<tr><th></th><th>∈</th><th>a</th><th>b</th><th>c</th></tr>
<tr><td>→p</td><td>∅</td><td>{p}</td><td>{q}</td><td>{r}</td></tr>
<tr><td>q</td><td>{p}</td><td>{q}</td><td>{r}</td><td>∅</td></tr>
<tr><td>*r</td><td>{q}</td><td>{r}</td><td>∅</td><td>{p}</td></tr>
</table></td><td>L3</td><td>CO2</td><td>5</td></tr>
<tr><td>c</td><td>Describe the four components of a grammatical description of a language.</td><td>L3</td><td>CO3</td><td>3</td></tr>
<tr><td>2.a</td><td>Design DFA accepting the following language for the alphabet {0,1}.
The set of all strings beginning with a '1' that interpreted as binary integer is a multiple of five.</td><td>L3</td><td>CO1</td><td>5</td></tr>
<tr><td>b</td><td>Compute $R_{12}^{(2)}$ using Kleene's theorem. 
</td><td>L3</td><td>CO2</td><td>5</td></tr>
<tr><td>c</td><td>Define the lemma that is used to prove that a given language is not regular.</td><td>L3</td><td>CO3</td><td>3</td></tr>
<tr><td>3.a</td><td>Given a Langague L = { awa : w ∈ {a,b}*}, Design a DFA to accept the language L^2.</td><td>L3</td><td>CO1</td><td>5</td></tr>
<tr><td>b</td><td>Consider the language L over {a,b} which contains the strings whose lengths are from the arithmetic progression.
P = {2,5,8,11....} = { 2 + 3n , n>=0 } That is, L = {x ∈ {a,b}*, |x| ∈ P }. Construct a DFA accepting L.</td><td>L2</td><td>CO2</td><td>5</td></tr>
<tr><td>c</td><td>Give the set description of the language of DFA and NFA.</td><td>L2</td><td>CO3</td><td>3</td></tr>
</table>

RAMAIAH
Institute of Technology

Department of Artificial Intelligence and Machine Learning

Programme: B E – Department of Artificial Intelligence and Machine Learning Internal Assessment – I

Instructions to Candidates: **answer any two**

Marks: 24

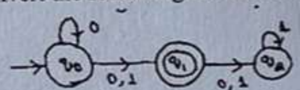
Q. NO	Questions	Marks
1.a	Differentiate between DFA and NFA. Design a DFA to accept strings of a's and b's not containing the substring abb.	4 4
b	Convert regular expression to NFA with e-transitions $(0+1)^*\ 10\ (0+1)$	
c	1. Give context-free grammars that generate the following language { $w \in \{0, 1\}^*$ | w contains at least three 1s } 2. Design a regular expression for the following i) $L = \{a^n b^m \mid n \geq 4 \text{ and } m \leq 3\}$ ii) $L = \{a^{2n} b^{2m} \mid n \geq 0, m \geq 0\}$.	4
2.a	Construct a DFA to accept strings over {a, b}, such that every block of length five at least two a's. Use extended transition of function to trace a string W = aabba. ((	4
b	Convert the following to NFA to its equivalent DFA 	4
c	1. Prove that the language $L = \left\{0^m 1^n \mid m > n, \Sigma = \{0,1\}\right\}$ is not regular. 2. Give context-free grammars that generate the following language { $w \in \{0, 1\}^*$ | $w = w^R$ and |w| is even }	4
3.a	Solve the following using Table Filling Theorem. (table below)	5
b	Define CFG. Design a context free grammar for the languages: i) $L = \{a^i b^j c^k$, where $i = j + k,\ i, j, k \geq 0\}$. ii) $L = \{0^{n+2}\ 1^n : n \geq 1\}$.	4
c	Mention applications of regular expression.	3

Table for 3.a:

δ	0	1
→A	B	E
B	C	F
* C	D	H
D	E	H
E	F	I
* F	G	B
G	H	B
H	I	C
* I	A	E

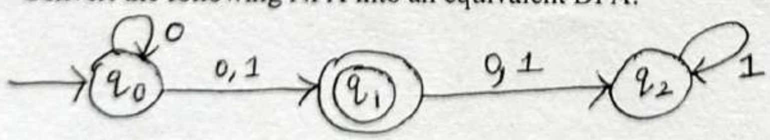
Department of Artificial Intelligence and Machine Lea

Programme: B.E | **Term:** May-August 2023 | **Max Marks:** 30
CIE: Test 1 | **Subject:** Theory of Computations | **Subject Code:** AI44
Credits: 2 :1 :0
Sem:IV | **Date: 12/06/2023** | **Time: 9:30 -10:30 am**

Portions for Test: (L1-L15)

Instructions to Candidates:

1. **Question 1 is Compulsory.** Answer any one question from 2 and 3.
2. Each Question carries 15M. **Write figures wherever necessary.**
3. Mobiles/Smart Gadgets are restricted.

<table>
<tr><th>Sl#</th><th>Question</th><th>Marks</th><th>Bloom's Level</th><th>CO Mapping</th></tr>
<tr><td>1</td><td>a. Design DFAs to accept the following languages.
i. String beginning with 'bab' over ∑= {a,b}.
ii. String not ending with 011 over ∑= {0,1}.</td><td>05</td><td>Apply</td><td>CO2</td></tr>
<tr><td></td><td>b. Convert the following NFA to a DFA over ∑= {a,b}.
<table>
<tr><th>δ NFA</th><th>ϵ</th><th>a</th><th>b</th></tr>
<tr><td>→ Q0</td><td>φ</td><td>φ</td><td>Q1</td></tr>
<tr><td>* Q1</td><td>Q2</td><td>Q1</td><td>Q2</td></tr>
<tr><td>Q2</td><td>φ</td><td>φ</td><td>Q0</td></tr>
</table></td><td>05</td><td>Apply</td><td>CO3</td></tr>
<tr><td></td><td>c. Consider the DFA.
<table>
<tr><th></th><th>0</th><th>1</th></tr>
<tr><td>→A</td><td>B</td><td>A</td></tr>
<tr><td>B</td><td>A</td><td>C</td></tr>
<tr><td>C</td><td>D</td><td>B</td></tr>
<tr><td>*D</td><td>D</td><td>A</td></tr>
<tr><td>E</td><td>D</td><td>F</td></tr>
<tr><td>F</td><td>G</td><td>E</td></tr>
<tr><td>G</td><td>F</td><td>G</td></tr>
<tr><td>H</td><td>C</td><td>D</td></tr>
</table>
i. Identify all distinguishable and indistinguishable pairs.
ii. Construct the minimum state equivalent DFA.</td><td>05</td><td>Apply</td><td>CO3</td></tr>
<tr><td>2</td><td>a. Differentiate between DFA, NFA and epsilon- NFA</td><td>05</td><td>Analyze</td><td>CO3</td></tr>
<tr><td></td><td>b. Show that the language L consisting of all palindromes over (0+1)* is not regular</td><td>05</td><td>Apply</td><td>CO3</td></tr>
<tr><td></td><td>c. Give DFA accepting the following language for the alphabet 0,1. Such that: the set of all strings beginning with a '1' that intercepted as binary integer is a multiple of 5.</td><td>05</td><td>Apply</td><td>CO2</td></tr>
<tr><td>3</td><td>a. State and prove pumping lemma for regular languages.</td><td>05</td><td>Understand</td><td>CO3</td></tr>
<tr><td></td><td>b. Convert the following NFA into an equivalent DFA.

</td><td>05</td><td>Apply</td><td>CO2</td></tr>
<tr><td></td><td>c. Define the following with an example each.
a) Language of DFA. b) Language of NFA.
c) Power of an Alphabet.
d) If M is a DFSM and ε ∈ L(M), what simple property must be true of M?
e) How Many Start and Final / Accepting a Finite Automata can have?</td><td>05</td><td>Remember</td><td>CO1</td></tr>
</table>

Course Outcomes meant to be assessed by the IA Test 1:

CO1: Explain the basic concepts of formal languages and finite automata. (PO-1,PSO-1)

CO2: Construct automata to accept strings from a specified language. (PO-1,2,3, PSO-1)

CO3: Convert among equivalently powerful notations for a language, including among DFAs, NFAs, and regular expressions, between PDAs, CFGs and normal forms of CFGs. (PO-1,2,3, PSO-1)

Internal Assessment Question Paper – 2
Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Artificial Intelligence and Data Science / Artificial Intelligence and Machine Learning

Programme: B.E		**Term:** May-August 2023	**Max Marks:** 30
CIE: Test 2	**Subject:** Theory of Computations	**Subject Code:** AI44/AD44	**Crédits:** 2 :1 :0
Sem:IV	**Date:** 27/07/2023	**Time:** 9:30 -10:30	**Portions for Test:** (L16-L28)

Instructions to Candidates:

1. **Question 1 is Compulsory.** Answer any one question from 2 and 3.
2. Each Question carries 15M. **Write figures wherever necessary.**
3. Mobiles/Smart Gadgets are restricted.

Sl#	Question	Marks	Bloom's Level	CO Mapping
1	a. Design PDA of the Language $L = \{ww^R \mid w \text{ is in } (0+1)^*\}$	05	Apply	CO 3
	b. Begin with the grammar: $S \rightarrow ABC \mid BaB$ $A \rightarrow aA \mid BaC \mid aaa$ $B \rightarrow bBb \mid a \mid D$ $C \rightarrow CA \mid AC$ $D \rightarrow \varepsilon$ i. Eliminate ε – productions. ii. Eliminate any unit productions in the resulting grammar. iii. Eliminate any useless symbols in the resulting grammar. iv. Put the resulting grammar into Chomsky Normal Form.	10	Apply	CO 4
2a	i. Design CFG for the language of palindromes over {0,1}. ii. Design CFG over {a, b} consisting of all strings of even length. iii. Abbreviate and define CFG.	05	Analyze	CO 3
b	State and prove pumping lemma for CFL.	05	Apply	CO 4
c	Design Turing Machine for the following language $L = \{a^n b^n c^n, n >= 1\}$, Also draw Transition diagram for the TM.	05	Apply	CO 5
3a	Given an instance of PCP, tell whether this instance has a solution. (see table below)	05	Understand	CO 5
b	Consider the grammar $S \rightarrow SbS \mid a$ This grammar is ambiguous; show in particular that the string abababa has two. i. Parse trees ii. Leftmost derivations iii. Rightmost derivations.	05	Apply	CO 3
c	Define Turing Machine with neat diagram of its abstract Structure. What is the language accepted by Turing machine?	05	Remember	CO 5

	List A	List B
i	w_i	x_i
1	110	110110
2	0011	00
3	0110	110

Course Outcomes meant to be assessed by the IA Test 2:

CO3: Convert among equivalently powerful notations for a language, CFGs and Normal forms of CFGs.

CO4: Prove the various closure and decision properties of formal languages.

CO5: Explain the concepts of Turing Machines, Post Correspondence problem.